# Technical Service Bulletin

| **Technical Service Bulletin:** tsb090029 | **Released Date:** 21-Apr-2009 |
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| **Injector Copper Seal Corrosion Issue** | |

# Injector Copper Seal Corrosion Issue

## Core Issue

The purpose of this Early Field Notification is to inform the field that QSK45 and QSK60 series injector copper seals could possibly become corroded if high sulfur fuel (>300 ppm) is used and either of the operating conditions listed below are present:

1. Engine is operating more than 50 percent of the time at low idle (and load factor <25 percent).
2. Ambient temperatures are below 5°C [41°F] for extended periods of time.

## Confirmation

QSK45 and QSK60 series engines with the HPI fuel system.

Sulfur corrosion has commonly been seen on QSK45 and QSK60 engines, but the potential exists for all QSK HPI engines to show signs of copper seal corrosion if the conditions listed above are met.

Copper seal corrosion is often referred to as a burned copper seal, due to the black marks found around the corroded or removed area.

Other symptoms could possibly include low cylinder temperatures or low power. Fault Code 621 (Exhaust Gas Temperature Deviation Low for Cylinder 1 - Data Valid But Below Normal Operational Range - Least Severe Level and Fault Code 622 ( Exhaust Gas Temperature Deviation Low for Cylinder 3 - Data Valid But Below Normal Operational Range - Least Severe Level) have also been seen. Most failures, however, are seen at engine rebuild or during tear down for another failure.

Injector copper seal corrosion can be identified by noting that a portion of the copper seal appears to be dissolved or worn away.

Injector copper seals could possibly become corroded if high sulfur fuel (>300 ppm) is used and either of the operating conditions listed below are present:

1. Engine is operating more than 50 percent of the time at low idle (and load factor <25 percent).
2. Ambient temperatures are below 5°C [41°F] for extended periods of time.

- If the engine is operating within the boundaries suggested above, it is recommended that routine inspection of one or more injector seals be carried out at approximate 6 month intervals. Replacement of the fuel injector seal set is recommended if corrosion is found.

Injector copper seal failures are normally seen at engine tear down due to another failure or during engine rebuild. The failure does **not** normally take the engine out of service, but it does increase the cost of the repair.

## Resolution

Replace corroded or burned copper seal with new seals (see figure 1). Analyze the sulfur content of fuel to make sure it is below 300 ppm. Analyze engine operation to make sure that the engine is not operating at low idle more than 50 percent (and load factor <25 percent) of the time. If ambient conditions have been below 5°C [41°F] and fuel with a >300 ppm sulfur content has been, used this could possibly also be an indication that copper seal damage is present. If repeated failures of copper seals are witnessed, it is recommended that the potential causes noted in this Early Field Notification be addressed for improved copper seal performance.


Figure 1. Illustration of a burned copper seal.

## Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

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